

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 12: Infimal Convolution

Based on Bauschke & Combettes

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Chapter Outline

- 1 Definition and Basic Facts
- 2 Infimal Convolution of Convex Functions
- 3 Moreau Envelope and Proximity Operator
- 4 Pasch–Hausdorff Envelope
- 5 Infimal Postcomposition
- 6 Summary and Applications

Notation You'll Need I

Throughout, $\mathcal{H}, \mathcal{K}$ are real Hilbert spaces with inner product $\langle \cdot | \cdot \rangle$ and induced norm $\|\cdot\|$.

- $\Gamma_0(\mathcal{H})$: proper, lower semicontinuous, convex functions $\mathcal{H} \rightarrow]-\infty, +\infty]$.
- $\text{dom } f = \{x \in \mathcal{H} : f(x) < +\infty\}$; f **proper** if $\text{dom } f \neq \emptyset$ and never takes value $-\infty$.
- $\text{epi } f = \{(x, \xi) \in \mathcal{H} \times \mathbb{R} : f(x) \leq \xi\}$, the **epigraph**.
- ι_C : **indicator function** (0 on C , $+\infty$ off C).
- $d_C(x) = \inf_{y \in C} \|x - y\|$: **distance** to C ; P_C : **projection** onto closed convex C .
- $\tau_y f = f(\cdot - y)$: **translation** by y .
- f^* : the **conjugate** (Fenchel transform) of f .

Notation You'll Need II

Key convention: Throughout, we write $\inf_{y \in \mathcal{H}}$ to mean the infimum is taken over the entire Hilbert space $\mathcal{H}$.

Definition 12.1: Infimal Convolution I

Definition 12.1 – For $f, g : \mathcal{H} \rightarrow]-\infty, +\infty]$:

The *infimal convolution* (or epigraphical sum) is

$$(f \square g) : \mathcal{H} \rightarrow]-\infty, +\infty] : x \mapsto \inf_{y \in \mathcal{H}} (f(y) + g(x - y)) \quad (1)$$

- $(f \square g)(x)$ is **exact at** x if the infimum is **attained** ($\exists y$ achieving the minimum).
- $(f \square g)$ is **exact** if exact at every point in its domain.
- When exact, we write $f \overline{\square} g$.

Definition 12.1: Infimal Convolution II

Key Properties from Definition:

(i) **Domain:**

$$\text{dom}(f \square g) = \text{dom } f + \text{dom } g = \{x + y : x \in \text{dom } f, y \in \text{dom } g\}$$

(ii) **Epigraph relation:**

$$\text{epi}(f \square g) = \{(x, \xi_1 + \xi_2) : x = y + z, (y, \xi_1) \in \text{epi } f, (z, \xi_2) \in \text{epi } g\}$$

(iii) **Commutative:** $f \square g = g \square f$ (when both sides defined).

(iv) **Associative:** $(f \square g) \square h = f \square (g \square h)$.

Example 12.2: Infimal Convolution of Indicator Functions

Example 12.2 – Let $C, D \subseteq \mathcal{H}$ be nonempty closed convex sets.

Then:

$$(\iota_C \square \iota_D)(x) = \iota_{C+D}(x) = \begin{cases} 0 & \text{if } x \in C + D \\ +\infty & \text{otherwise} \end{cases}$$

Interpretation: Infimal convolution of indicator functions yields the indicator of the Minkowski sum.

General composition rule:

$$\iota_C \square f = f|_C \quad \text{and} \quad f \square \iota_C = f(-\cdot)|_{-C}$$

Definition 12.5: Continuous Affine Minorant

Definition 12.5 – f possesses a *continuous affine minorant with slope* $u \in \mathcal{H}$ if

$$\langle \cdot \mid u \rangle \text{ is bounded below by } f$$

That is, $\exists \eta \in \mathbb{R}$ such that $f(y) \geq \langle y \mid u \rangle + \eta$ for all $y \in \mathcal{H}$.

Intuition: The linear functional with slope u provides a lower bound for f .

Examples:

- If f is differentiable with bounded gradient, it has a continuous affine minorant.
- The absolute value $f(x) = |x|$ has affine minorant with any $|u| \leq 1$.

Proposition 12.6: Properties with Affine Minorants I

Proposition 12.6 – Suppose $f, g, h : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $u \in \mathcal{H}$.

- (i) If f and g possess continuous affine minorants with slope u , then $f \square g$ possesses such a minorant, and $-\infty \notin (f \square g)(\mathcal{H})$.
- (ii) $\text{dom}(f \square g) = \text{dom } f + \text{dom } g$.
- (iii) $f \square g = g \square f$ (commutativity).
- (iv) Suppose f, g, h all possess affine minorants with the same slope. Then $(f \square g) \square h = f \square (g \square h)$ (associativity).

Proposition 12.6: Properties with Affine Minorants II

Key Formula: From the definition,

$$(f \square g)(x) = \inf_{(y,z) \in \mathcal{H} \times \mathcal{H}, y+z=x} (f(y) + g(z))$$

Remark 12.7: The conditions on affine minorants are essential. Without them, pathological behavior can occur (e.g., $g \equiv -\infty$).

Proposition 12.8: Epigraph and Exactness

Proposition 12.8 – For functions $f, g : \mathcal{H} \rightarrow]-\infty, +\infty]$:

- (i) $\text{epi } f + \text{epi } g \subseteq \text{epi}(f \square g)$.
- (ii) If $f \square g = f \bar{\square} g$ (is exact), then

$$\text{epi } f + \text{epi } g = \text{epi}(f \square g)$$

Interpretation: The epigraph of an infimal convolution contains the Minkowski sum of the epigraphs. Equality holds precisely when the infimal convolution is exact.

Impact: This connects the algebraic operation (infimal convolution) to geometric properties (epigraphs and their sums).

Proposition 12.9: Infimal Convolution with Power of Norm

Proposition 12.9 – Let f be proper, $p \in [1, +\infty[$, and define

$$g_\gamma = f \square \left(\frac{1}{\gamma p} \|\cdot\|^p \right)$$

Then for all $\gamma \in \mathbb{R}_{++}$:

- (i) $\text{dom } g_\gamma = \mathcal{H}$.
- (ii) $\inf f(\mathcal{H}) \leq g_\gamma(x) \leq g_\mu(x) \leq f(x)$ when $0 < \gamma < \mu$.
- (iii) $\inf g_\gamma(\mathcal{H}) = \inf f(\mathcal{H})$.
- (iv) $g_\gamma(x) \downarrow \inf f(\mathcal{H})$ as $\gamma \rightarrow +\infty$.
- (v) g_γ is bounded above on every ball in $\mathcal{H}$.

Formula:

$$g_\gamma(x) = \inf_{y \in \mathcal{H}} \left(f(y) + \frac{1}{\gamma p} \|x - y\|^p \right)$$

Proposition 12.11: Infimal Convolution of Convex Functions

Proposition 12.11 – If f and g are convex functions from $\mathcal{H}$ to $] -\infty, +\infty]$, then $f \square g$ is convex.

Proof outline: Define the marginal function

$$F : \mathcal{H} \times \mathcal{H} \rightarrow] -\infty, +\infty], \quad F(y, z) = f(y) + g(z)$$

F is convex (product of convex functions). The infimal convolution corresponds to the marginal function

$$(f \square g)(x) = \inf_{y+z=x} F(y, z)$$

By Proposition 8.35, the marginal of a convex function is convex. $\square$

Key consequence: Many important functions (Moreau envelope, distance to convex set, etc.) inherit convexity via infimal convolution.

Corollary 12.12: Distance Function to Convex Set

Corollary 12.12 – Let C be a convex subset of $\mathcal{H}$. Then the distance function

$$d_C(x) = \inf_{y \in C} \|x - y\|$$

is convex.

Proof: By Example 12.2,

$$d_C = \iota_C \square \|\cdot\|$$

Both ι_C and $\|\cdot\|$ are convex, so their infimal convolution is convex. $\square$

Remark: This is a powerful tool: convexity of the distance function ensures nice properties in optimization.

Proposition 12.14: Exactness for Convex Functions I

Proposition 12.14 – Let $f, g \in \Gamma_0(\mathcal{H})$, and suppose one:

- (i) f is **supercoercive**, or
- (ii) f is **coercive** and g is **bounded below**

Then $f \square g = f \overline{\square} g \in \Gamma_0(\mathcal{H})$, i.e., the infimal convolution is exact and returns to Γ_0 .

Key idea of proof:

If $(x_n) \rightarrow x$ and $(f \square g)(x_n) \rightarrow \mu$, then there exist minimizers y_n such that $f(y_n) + g(x_n - y_n) = (f \square g)(x_n)$.

By supercoercivity or coercivity, (y_n) remains bounded. Passing to a subsequence and using lower semicontinuity shows

$$(f \square g)(x) \leq \liminf_n (f \square g)(x_n) = \mu$$

Thus, $f \square g$ is sequentially lower semicontinuous.

Proposition 12.14: Exactness for Convex Functions II

Consequence: Under mild conditions, the infimal convolution of two proper convex lsc functions is again proper convex lsc.

Definition 12.20: Moreau Envelope

Definition 12.20 – Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $\gamma \in \mathbb{R}_{++}$.
The *Moreau envelope* of f with parameter γ is:

$$\gamma f = f \square \left(\frac{1}{2\gamma} \|\cdot\|^2 \right) \quad (2)$$

$$: \mathcal{H} \rightarrow]-\infty, +\infty] \quad (3)$$

$$: x \mapsto \inf_{y \in \mathcal{H}} \left(f(y) + \frac{1}{2\gamma} \|x - y\|^2 \right) \quad (4)$$

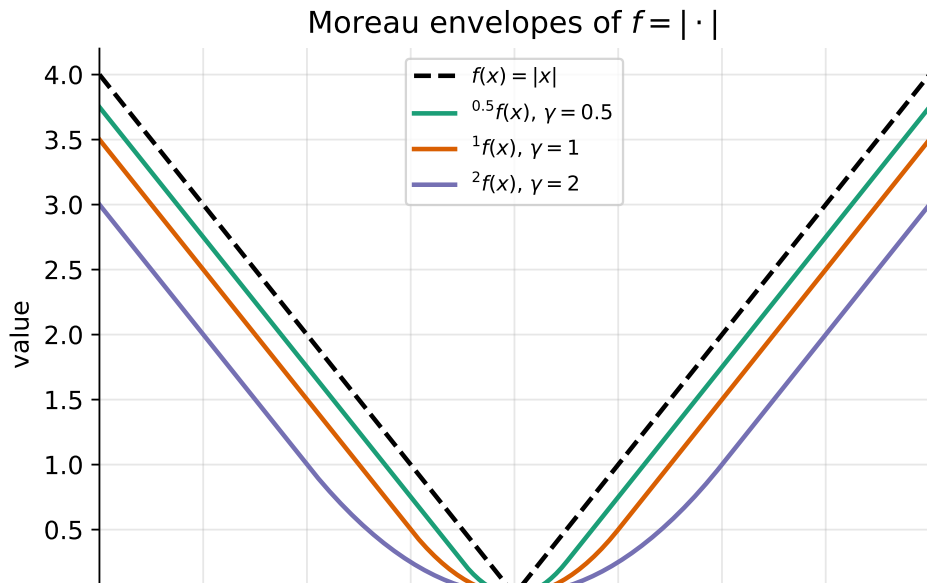
Alternative notation: f_γ , $J_\gamma f$, $M_\gamma f$.

Example 12.21: For a convex set C ,

$$\gamma f_{\iota_C} = \frac{1}{2\gamma} d_C^2$$

The Moreau envelope of an indicator is the (normalized) squared distance.

Figure: Moreau Envelope



Proposition 12.22: Composition Property I

Proposition 12.22 – For $\gamma, \mu \in \mathbb{R}_{++}$:

(i) $\mu(\gamma f) = \gamma f(\mu f)$.

(ii) $\gamma f(\mu f) = \gamma + \mu f$.

Proof of (ii): Unfold the definitions:

$$\gamma f(\mu f) = \left(f \square \frac{1}{2\mu} \|\cdot\|^2 \right) \square \frac{1}{2\gamma} \|\cdot\|^2 \quad (5)$$

$$= f \square \left(\frac{1}{2\mu} \|\cdot\|^2 \square \frac{1}{2\gamma} \|\cdot\|^2 \right) \quad (6)$$

$$= f \square \frac{1}{2(\gamma + \mu)} \|\cdot\|^2 \quad (7)$$

$$= \gamma + \mu f \quad (8)$$

The middle step uses a classical identity on composition of the squared-norm function.

Proposition 12.22: Composition Property II

Significance: The Moreau envelope is stable under composition. Applying the Moreau envelope multiple times with parameters γ_1, γ_2 is equivalent to applying once with $\gamma_1 + \gamma_2$.

Proposition 12.30: Differentiability of Moreau Envelope

Proposition 12.30 – Let $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \mathbb{R}_{++}$. Then $\gamma f : \mathcal{H} \rightarrow \mathbb{R}$ is Fréchet differentiable on $\mathcal{H}$, with gradient

$$\nabla(\gamma f) = \gamma^{-1}(\text{Id} - \text{Prox}_{\gamma f})$$

where $\text{Prox}_{\gamma f}$ is the proximity operator (see next).

Moreover, $\nabla(\gamma f)$ is γ^{-1} -Lipschitz continuous.

Key implication: The Moreau envelope provides a smooth approximation to a non-smooth function f .

Application: In optimization, one can use smooth gradient methods on γf instead of non-smooth algorithms on f .

Definition 12.23: Proximity Operator

Definition 12.23 – Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $x \in \mathcal{H}$.

The unique point $p \in \mathcal{H}$ satisfying

$$f(x) = \min_{y \in \mathcal{H}} \left(f(y) + \frac{1}{2} \|x - y\|^2 \right) = f(p) + \frac{1}{2} \|x - p\|^2$$

is called the *proximity operator* or *proximal mapping* of f at x :

$$p = \text{Prox}_f(x)$$

Alternative notation: $\text{prox}_f(x)$, $J_f(x)$.

Optimality condition:

$$x - \text{Prox}_f(x) \in \partial f(\text{Prox}_f(x))$$

(See Chapter 14 on subdifferentials.)

Proposition 12.26: Characterization I

Proposition 12.26 – Let $f \in \Gamma_0(\mathcal{H})$. Then $p = \text{Prox}_f(x)$ if and only if

$$\forall y \in \mathcal{H}, \quad \langle y - p \mid x - p \rangle + f(p) \leq f(y)$$

Interpretation: This is a variational characterization: p is the point that minimizes $y \mapsto f(y) + \frac{1}{2} \|x - y\|^2$, and the above inequality captures the first-order optimality.

Proposition 12.26: Characterization II

Remark 12.24: From Proposition 12.22(i), the Moreau envelope can be written as

$$\gamma f(x) = f(\text{Prox}_{\gamma f} x) + \frac{1}{2\gamma} \|x - \text{Prox}_{\gamma f} x\|^2$$

This shows that the Moreau envelope is determined by the proximity operator.

Example 12.25: Projection onto Convex Set

Example 12.25 – If C is a nonempty closed convex set, then

$$\text{Prox}_{\iota_C} = P_C$$

The proximity operator of the indicator function is orthogonal projection onto C .

Proof:

$$\text{Prox}_{\iota_C}(x) = \arg \min_{y \in \mathcal{H}} \left(\iota_C(y) + \frac{1}{2} \|x - y\|^2 \right) \quad (9)$$

$$= \arg \min_{y \in C} \|x - y\|^2 \quad (10)$$

$$= P_C(x) \quad (11)$$

Consequence: Projection is a special case of the proximity operator.

Soft-Thresholding: $\text{Prox}_{|\cdot|}$

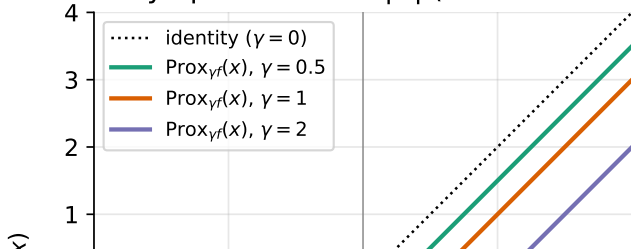
Fundamental example: For $f(x) = |x|$,

$$\text{Prox}_{\gamma|\cdot|}(x) = \text{sign}(x) \max(|x| - \gamma, 0)$$

This is the **soft-thresholding** operator, crucial in:

- Sparse optimization and compressed sensing.
- ℓ^1 -regularized problems.
- Signal and image processing.

Proximity operator of $f = |\cdot|$ (soft-thresholding)



Proposition 12.27-12.29: Nonexpansiveness and Fixed Points I

Proposition 12.28 – The operators Prox_f and $\text{Id} - \text{Prox}_f$ are *firmly nonexpansive*:

- (a) $\|\text{Prox}_f x - \text{Prox}_f y\| \leq \|x - y\|$ (nonexpansive).
- (b) $\|\text{Prox}_f x - \text{Prox}_f y\|^2 + \|(x - \text{Prox}_f x) - (y - \text{Prox}_f y)\|^2 \leq \|x - y\|^2$.

Proposition 12.29 – For any $f \in \Gamma_0(\mathcal{H})$:

$$\text{Fix}(\text{Prox}_f) = \text{Argmin } f$$

The fixed points of the proximity operator are exactly the minimizers of f .

Proposition 12.27-12.29: Nonexpansiveness and Fixed Points II

Significance for optimization:

- Firm nonexpansiveness ensures convergence of iterative schemes.
- Fixed-point characterization shows that finding minimizers reduces to finding fixed points of Prox_f .
- This underlies proximal gradient descent, Douglas–Rachford, ADMM, etc.

Corollary 12.31: Distance Function is Smooth

Corollary 12.31 – Let C be a nonempty closed convex set. Then d_C^2 is Fréchet differentiable on $\mathcal{H}$, with

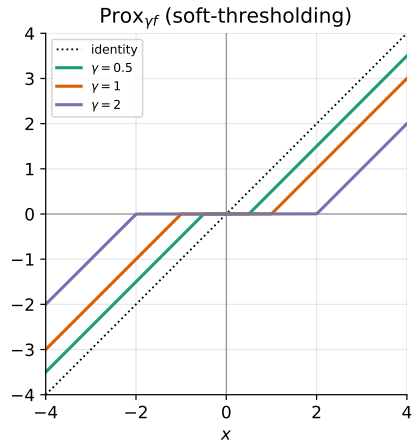
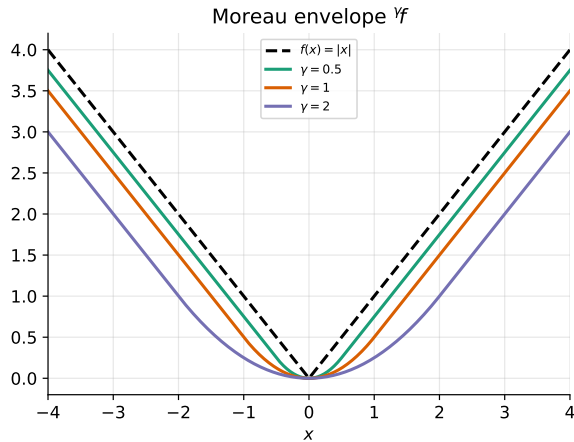
$$\nabla(d_C^2) = 2(\text{Id} - P_C)$$

Proof: Apply Proposition 12.30 with $f = \iota_C$ and $\gamma = 1/2$. By Example 12.25, $\text{Prox}_{\iota_C} = P_C$. Thus,

$$\nabla\left(\frac{1}{2}d_C^2\right) = 2(\text{Id} - P_C) \Rightarrow \nabla(d_C^2) = 2(\text{Id} - P_C)$$

Impact: Squared distance to a convex set is smooth, enabling the use of gradient-based optimization methods.

Figure: Moreau Envelope and Proximity Operator



Proposition 12.33: Convergence of Sequences

Proposition 12.33 – For $f \in \Gamma_0(\mathcal{H})$ and $x \in \mathcal{H}$:

The nets $(\gamma f(x))_{\gamma \in \mathbb{R}_{++}}$ and $(f(\text{Prox}_{\gamma f} x))_{\gamma \in \mathbb{R}_{++}}$ are decreasing with:

- (i) $\gamma f(x) \downarrow \inf f(\mathcal{H})$ as $\gamma \rightarrow +\infty$.
- (ii) $\gamma f(x) \uparrow f(x)$ as $\gamma \downarrow 0$.
- (iii) If $x \in \text{dom } f$, then $\gamma f(x) \rightarrow f(x)$ and $\gamma^{-1} \|x - \text{Prox}_{\gamma f} x\|^2 \rightarrow 0$ as $\gamma \downarrow 0$.

Interpretation:

- As $\gamma \rightarrow 0$: Moreau envelope recovers the original function.
- As $\gamma \rightarrow \infty$: Envelope becomes flat, approaching the infimum.
- Proximity points converge to the set of minimizers.

Definition 12.16: Pasch–Hausdorff Envelope

Definition 12.16 – Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $\beta \in \mathbb{R}_{++}$.
The β -Pasch–Hausdorff envelope of f is:

$$f \square (\beta \|\cdot\|)$$

This is the infimal convolution of f with β times the norm.

Proposition 12.17: Lipschitz Characterization I

Proposition 12.17 – Let f be proper, $\beta \in \mathbb{R}_{++}$, and $g = f \square (\beta \|\cdot\|)$ (the Pasch–Hausdorff envelope). Then exactly one holds:

- (i) f possesses a β -Lipschitz continuous minorant. In this case, g is the largest such minorant.
- (ii) f has no β -Lipschitz continuous minorant. In this case, $g \equiv -\infty$.

Significance: The Pasch–Hausdorff envelope computes the largest β -Lipschitz lower bound for f .

Proposition 12.17: Lipschitz Characterization II

Corollary 12.18 – If f is β -Lipschitz continuous, then

$$f \square (\beta \|\cdot\|) = f$$

A Lipschitz function is its own envelope.

Corollary 12.19 – For a function $h : C \rightarrow \mathbb{R}$ that is β -Lipschitz on C , the function

$$(f \square (\beta \|\cdot\|))|_{\mathcal{H}}$$

where $f = \iota_C \vee h$ (indicator/extension), extends h to all of $\mathcal{H}$ while preserving the β -Lipschitz property.

Definition 12.34: Infimal Postcomposition

Definition 12.34 – For $f : \mathcal{H} \rightarrow]-\infty, +\infty]$, $L : \mathcal{H} \rightarrow \mathcal{K}$, and $y \in \mathcal{K}$:
The *infimal postcomposition* of f by L is:

$$(L \triangleright f)(y) := \inf_{\substack{x \in \mathcal{H} \\ Lx=y}} f(x)$$

Compare:

- **Infimal convolution:** $(f \square g)(x) = \inf_y [f(y) + g(x - y)]$.
- **Infimal postcomposition:** $(L \triangleright f)(y) = \inf_{Lx=y} f(x)$.

$(L \triangleright f)$ is exact at y if the minimum is attained.

Proposition 12.36-12.37: Properties I

Proposition 12.36 – For $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ and $L : \mathcal{H} \rightarrow \mathcal{K}$:

- (i) $\text{dom}(L \triangleright f) = L(\text{dom } f)$.
- (ii) If f is convex and L is affine, then $L \triangleright f$ is convex.

Example 12.35: Latent Group Lasso

In machine learning, given index sets $I_1, \dots, I_K$ partitioning $\{1, \dots, m\}$, the group lasso penalty is

$$f(x) = \sum_{k=1}^K \|x_{I_k}\|_p$$

The latent (or Group) Lasso norm is the infimal postcomposition with the summation map.

Proposition 12.36-12.37: Properties II

Proposition 12.37 – For $f, g : \mathcal{H} \rightarrow]-\infty, +\infty]$:

$$f \square g = L \triangleright (f \oplus g)$$

where $L : (y, z) \mapsto y + z$ is summation, and $f \oplus g(y, z) = f(y) + g(z)$.

Interpretation: Infimal convolution decomposes as infimal postcomposition of a direct sum.

Key Takeaways

- (1) **Infimal convolution** $(f \square g)(x) = \inf_y [f(y) + g(x - y)]$ is the fundamental operation connecting Γ_0 functions.
- (2) **Moreau envelope** $\gamma f = f \square \frac{1}{2\gamma} \|\cdot\|^2$ provides a smooth approximation of f .
- (3) **Proximity operator** Prox_f is the minimizer in the Moreau definition and generalizes orthogonal projection.
- (4) **Pasch–Hausdorff envelope** $f \square (\beta \|\cdot\|)$ gives the largest β -Lipschitz lower bound.
- (5) **Infimal postcomposition** $(L \triangleright f)(y) = \inf_{Lx=y} f(x)$ minimizes over level sets.
- (6) These are central to modern first-order optimization algorithms.

Applications to Optimization I

Proximal Methods: The proximity operator is the workhorse of first-order optimization.

① **Proximal gradient descent:**

$$x^{k+1} = \text{Prox}_{\gamma f}(x^k - \gamma \nabla g(x^k))$$

for minimizing $f + g$ where g is smooth.

② **Douglas–Rachford splitting:** Alternating proximal steps for $\min(f + g)$:

$$y^{k+1} = \text{Prox}_{\gamma g}(x^k), \quad x^{k+1} = \text{Prox}_{\gamma f}(2y^{k+1} - x^k)$$

③ **ADMM (Alternating Direction Method of Multipliers):** Distributed/parallel first-order algorithm using proximal steps.

Applications to Optimization II

Smoothing: Use the Moreau envelope to smooth non-smooth functions.

Analysis: Properties like firm nonexpansiveness guarantee convergence.

Next Steps (Chapters 13–15):

- **Chapter 13:** Conjugacy (Fenchel transform) and duality.
- **Chapter 14:** Subdifferential calculus.
- **Chapter 15:** Applications to concrete problems.
- **Part IV:** Monotone operator theory and splitting algorithms.

Historical and Further Reading I

Classical foundations:

- **Moreau (1962):** Introduced the Moreau envelope and proximity operator.
- **Rockafellar (1976):** Developed the proximal point algorithm and monotone operator theory.

Modern references:

- Bauschke, H. H., & Combettes, P. L. (2017). *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed. Springer.
- Nesterov, Y. (2004). *Introductory Lectures on Convex Optimization*. Kluwer.
- Boyd, S., Boyd, S. P., & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.

Historical and Further Reading II

Application areas:

- Signal and image processing.
- Machine learning and sparse optimization.
- Optimal transport and computational geometry.
- Control theory and game theory.

Key implementation libraries:

- `ProximalOperators.jl` (Julia).
- `ProximalAlgorithms.jl` (Julia).
- `sklearn.preprocessing` (Python, sparse/soft-threshold utilities).